Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm for 2.5D Chiplet Networks

Yusuf Tahir Orhan - N24112049*
*Computer Engineering Department, Hacettepe University, Ankara, Türkiye
†Graduate School of Science and Engineering, Hacettepe University, Ankara, Türkiye
E-mails: *yusuftahirorhan@hacettepe.edu.tr

***Abstract*—While 2.5-dimensional (2.5D) chiplet integration offers a scalable and cost-effective solution for modern modular systems, it introduces critical routing challenges, including system-wide deadlocks and high susceptibility to Vertical Link (VL) failures. Existing fault-tolerant routing techniques designed for traditional two-dimensional (2D) and three-dimensional (3D) Networks-on-Chip (NoC) are inadequate for the unique traffic patterns and irregular topologies of 2.5D architectures. To address these limitations, this project implements the Deadlock-Free and Fault-Tolerant (DeFT) routing algorithm as a resilient baseline for inter-chiplet communication. The methodology employs a rule-based Virtual Network (VN) assignment strategy to guarantee deadlock-freedom, coupled with a dynamic, congestion-aware VL selection mechanism to maintain connectivity despite hardware failures. The system architecture, comprising multiple chiplets on an active interposer, is modeled by extending the cycle-accurate Noxim simulator. To validate the implementation, the algorithm will be evaluated under various traffic models and progressive VL fault injection scenarios. By analyzing network reachability, latency, and throughput, this study aims to demonstrate the algorithm's capability to tolerate severe fault rates, ultimately establishing a robust foundation for future reinforcement learning-based routing optimizations.**

***Index Terms*—2.5D integration, chiplet networks, Network-on-Chip (NoC), deadlock-free routing, fault tolerance.**

I. Introduction

The continuous demand for higher computation power necessitates further scalability improvements in modern hardware design. Conventional 2D Systems-on-Chip (SoC) suffer from high manufacturing costs and low yield as they scale to accommodate greater complexities. To address these limitations, 2.5D integration has emerged as a highly effective modular solution. In a typical 2.5D architecture, multiple smaller dies, known as *chiplets*, are integrated onto an active interposer. While the individual chiplets may rely on locally deadlock-free Networks-on-Chip (NoC), interconnecting them introduces severe routing challenges. Specifically, the "up-down-up" traffic pattern required to route packets between chiplets via the interposer can create cyclic dependencies, leading to system-wide deadlocks.

Furthermore, the reliability of these modular systems is heavily dependent on the Vertical Links (VLs) that connect the chiplets to the interposer. These VLs are highly susceptible to manufacturing defects, thermomigration, and electromigration. Existing fault-tolerant routing techniques designed for traditional 2D and 3D architectures cannot be directly applied to the unique structural constraints and irregular topologies of 2.5D networks. In standard systems, even a small number of VL failures can completely disrupt connectivity, rendering the hardware useless and severely degrading overall throughput and latency.

To address these critical gaps, this project implements and evaluates the DeFT (Deadlock-Free and Fault-Tolerant) routing algorithm [1]. The proposed approach utilizes a rule-based Virtual Network (VN) assignment strategy with two virtual channels to guarantee deadlock-freedom without artificially restricting VL selection. Alongside this, a dynamic, congestion-aware VL selection mechanism is employed to bypass faulty microbumps and maintain balanced network utilization.

The primary objective of this study is to extend the cycle-accurate Noxim simulator [2] to support 2.5D chiplet topologies and dynamic fault injection models. By systematically evaluating DeFT under various traffic patterns and fault rates (up to 25%), this project will analyze network reachability and latency improvements. Ultimately, this implementation will serve as a robust, fully reproducible baseline for ongoing research focused on developing reinforcement learning-based routing algorithms for 2.5D chiplet networks.

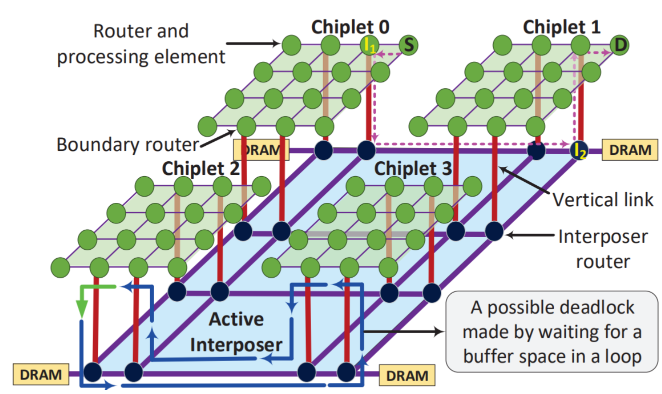

Fig. 1. An abstract overview of the baseline 2.5D network architecture featuring four chiplets on an active interposer, illustrating vertical links and a potential inter-chiplet deadlock scenario.

II. Related Work

The design of robust communication protocols for 2.5D integrated systems has evolved significantly, focusing on deadlock avoidance in modular topologies and fault-tolerant mechanisms for vertical interconnects.

Modular routing designs, such as the Modular Turn Restriction (MTR) [3], utilize restricted turn models to break cyclic dependencies across chiplet boundaries. However, more recent frameworks like the BUP (Update Packet Bypassing) mechanism [4] introduce a deadlock resolution framework specifically designed to handle dependencies by bypassing update packets, which offers higher flexibility than traditional turn restrictions.

In terms of reliability, the ReD (Reliable and Deadlock-Free) algorithm [5] represents the state-of-the-art in interposer-based systems, building upon the principles of Virtual Network (VN) separation to ensure both deadlock-freedom and resilience against Vertical Link (VL) failures. Reliability in multi-die systems is also extensively studied in 3D Networks-on-Chip (NoC). For instance, circular routing schemes [6] and runtime fault-tolerant routing for partially connected 3D architectures [7] provide insights into maintaining reachability under Through-Silicon Via (TSV) failures. While these 3D methods are effective, their application to 2.5D systems is constrained by the specific “up-down-up” traffic patterns of active interposers. Additionally, hardware-level solutions, such as configurable fault-tolerant links [8], emphasize the importance of link-level reconfiguration to bypass permanent defects.

This implementation of the DeFT algorithm [1] serves as a high-fidelity baseline by integrating these reliability principles. It provides the necessary foundation for my Master’s thesis, which aims to optimize these adaptive routing strategies using reinforcement learning.

TABLE I
Comparison of Routing and Reliability Frameworks

Work	Year	2.5D	Deadlock-Free	Fault-Tolerant	Adaptive	Metric
MTR [3]	2018	✓	✓	✗	✗	Latency
RC [9]	2020	✓	✓	✗	✓	Throughput
DeFT [1]	2022	✓	✓	✓	✓	Reachability
ReD [5]	2023	✓	✓	✓	✓	Reliability
BUP [4]	2023	✓	✓	✓	✗	Area
Salamat [7]	2018	✗(3D)	✓	✓	✓	Runtime

III. Problem Definition

The core problem addressed in this project is to guarantee deadlock-free and fault-tolerant inter-chiplet routing in a 2.5D integrated system without artificially restricting routing paths. This section defines the precise system parameters, evaluation models, and the mathematical formulation used for the dynamic routing logic.

A. System and Application Description

To ensure reproducibility, the experimental environment and application scenario are strictly defined under three operational models:

1) System Model: The target architecture is a 2.5D heterogeneous system comprising 4 CPU chiplets integrated onto an active interposer. Each chiplet utilizes a 4×4 2D-mesh Network-on-Chip (NoC) topology. Inter-chiplet communication is facilitated by 4 bidirectional Vertical Links (VLs) per chiplet, located at the boundaries of the local mesh. The routers directly attached to these VLs act as *boundary routers*, interfacing between the intra-chiplet network and the active interposer. The packet size is configured to 8 flits, with a buffer depth of 4 flits per Virtual Channel (VC).

2) Traffic Model: The evaluation framework incorporates both synthetic and real-world traffic patterns to comprehensively assess network behavior. Synthetic evaluations use Uniform, Localized (where 40% of traffic remains intra-chiplet), and Hotspot (3 hotspot nodes with a 10% injection rate) distributions. Real-application traffic is modeled using PARSEC benchmark traces (e.g., blackscholes, fluidanimate, swaptions) generated via the GEM5 full-system simulator.

3) Fault Model: The reliability aspect focuses on permanent defects in the microbumps connecting chiplets to the interposer. Faults are injected randomly into the 16 total VLs of the 4-chiplet system. The fault injection rate varies from 3.125% (1 faulty VL) to 25% (8 faulty VLs). The primary constraint of this model is that no single chiplet is entirely disconnected from the interposer (i.e., at least one VL per chiplet remains functional).

B. Objectives

The primary objectives of this implementation are as follows:

- **Deadlock Avoidance:** Ensure 100% deadlock-free routing for all inter-chiplet traffic using a Virtual Network (VN) separation technique with minimal hardware overhead.
- **Maximum Reachability:** Maintain 100% network reachability even under severe fault conditions (up to 25% VL failure rate).
- **Latency Minimization:** Improve average packet latency by dynamically balancing the traffic load across available VLs rather than relying on strict distance-based routing.

C. Constraints

The system design is bounded by several hardware and architectural constraints:

- **Memory/Hardware Constraint:** To keep the area and power overhead below 2%, the routing algorithm is strictly limited to using a maximum of two Virtual Channels (VCs) per physical port.
- **Topological Constraint:** Each chiplet is limited to exactly 4 VLs to represent a realistic, cost-effective manufacturing constraint rather than assuming a fully connected 3D TSV matrix.
- **Routing Constraint:** VN transitions (from VN.0 to VN.1) are strictly uni-directional to prevent cyclic dependencies between virtual networks.

D. Mathematical Formulation for VL Selection

Although this project is primarily an implementation study, the selection of the optimal Vertical Link (VL) in the presence of faults relies on an offline optimization metric. The objective is to select a set of VLs, s^*, for all routers on a source chiplet that minimizes both the traffic load variance and the physical distance.

Let R_C be the set of routers on chiplet C, and VL_C be the set of fault-free vertical links. If $U_r^v \in \{0,1\}$ denotes whether router r utilizes VL v, the traffic load l_v on a specific VL is defined as:

$$l_v = \sum_{r \in R_C} T_r^{inter} \times U_r^v \tag{1}$$

where T_r^{inter} is the inter-chiplet traffic injection rate of router r. The load cost L_v measures the deviation from the average VL load (l_{avg}):

$$L_v = \left| \frac{l_v - l_{avg}}{l_{avg}} \right| \tag{2}$$

Simultaneously, the physical distance (hop-count) D_v^r between a router $r(x_r, y_r)$ and a VL $v(x_v, y_v)$ is calculated using Manhattan distance:

$$D_v^r = |x_r - x_v| + |y_r - y_v| \tag{3}$$

The total distance cost D_v for a link v is the sum of distances from all routers utilizing it:

$$D_v = \sum_{r \in R_C} D_v^r \times U_r^v \tag{4}$$

The overall cost function C_s for a given selection set s is formulated as the weighted sum of distance and load costs:

$$C_s = \sum_{v \in VL_C} (\rho \times D_v) + L_v \tag{5}$$

where ρ is a weighting coefficient determining the trade-off between distance minimization and load balancing (set to $\rho = 0.01$ based on experimental heuristics). The goal of the algorithm is to find the assignment U_r^v that minimizes C_s.

TABLE II
MATHEMATICAL NOTATION FOR VL SELECTION FORMULATION

Symbol	Description
C	A specific chiplet in the network
R_C	Set of routers belonging to chiplet C
VL_C	Set of functional (fault-free) Vertical Links on C
T_r^{inter}	Inter-chiplet traffic injection rate of router r
U_r^v	Binary decision variable: 1 if router r uses VL v, 0 otherwise
l_v, l_{avg}	Traffic load on VL v and the average load across all VLs
L_v	Load imbalance cost for VL v
D_v^r, D_v	Hop distance from r to v, and total distance cost for v
ρ	Weighting coefficient for distance vs. load trade-off
C_s	Total cost of a specific VL assignment set s

IV. System Architecture and Design

This section details the system-level design of the proposed 2.5D Network-on-Chip (NoC) architecture, the hardware-software interaction within the simulation environment, and the computational models governing the routing logic. The high-level physical architecture was previously illustrated in Fig. 1.

A. Hardware and Software Architecture

The system architecture is bifurcated into the simulated physical hardware topology and the software logic controlling the network behavior within the Noxim framework [2].

Hardware Architecture (Simulated):

- **Processing Elements (PE) and Routers:** Each chiplet contains a 4×4 mesh of routers, each attached to a PE generating traffic.
- **Active Interposer:** A larger underlying mesh connecting the overhead chiplets.
- **Vertical Links (VLs):** 4 bidirectional interconnects per chiplet acting as physical bridges between the boundary routers of the chiplet and the interposer routers.

Software/Logic Architecture (Noxim Extensions):

- **DeFT Routing Logic:** Computes the deterministic minimal path and executes Virtual Network (VN) assignment.
- **Lookup Tables (LUT):** Stored within the boundary routers, these pre-computed tables represent the VL selection logic for different fault scenarios.
- **Fault Injection Manager:** A centralized software module that dynamically disables specific VL ports at the start of the simulation based on the defined fault rate (e.g., 12.5% or 25%).

B. Dataflow and System Model

Data flows through the 2.5D system in a packet-switched manner, utilizing an "up-down-up" trajectory for inter-chiplet communication. When a source PE generates a packet destined for a different chiplet, the dataflow is as follows: 1. The packet is assigned to VN.0 at the source router. 2. It travels through the intra-chiplet mesh to the designated boundary router (determined by the VL lookup table). 3. It traverses *down* the VL to the active interposer. 4. It travels across the interposer mesh to the target chiplet's footprint. 5. Between leaving the source and entering the destination chiplet, the packet may switch to VN.1 based on DeFT's deadlock-avoidance rules. 6. It traverses *up* the destination VL and routes locally to the final PE.

C. Requirements Analysis (Implementation Track)

1) Functional Requirements:

1) The system shall route inter-chiplet packets without deadlocks using exactly two Virtual Channels (VCs).
2) The simulator shall implement the three VN-assignment rules: (a) VN.1 to VN.0 is forbidden, (b) Up-to-Horizontal is forbidden in VN.0, and (c) Horizontal-to-Down is forbidden in VN.1.

3) Boundary routers shall read from local lookup tables to dynamically select intermediate VL destinations based on the current fault state.
4) The simulation framework shall support centralized injection of permanent VL faults before the routing stage begins.

2) Non-Functional Requirements:

- **Reliability:** The routing algorithm must maintain 100% network reachability with up to 25% of VLs disabled.
- **Overhead:** The area and power overhead introduced by the DeFT logic and lookup tables must remain minimal (conceptually $< 2\%$ as per literature [1]).
- **Latency:** The system must minimize average packet latency by utilizing load-balanced VL selection.

D. Timing and Execution Model

The system operates on a cycle-accurate, clock-driven timing model inherent to hardware NoCs. The execution model is structured around a standard 5-stage router pipeline:

- **Execution Model:** Clock-driven synchronous pipeline.
- **Timing Constraints:** Each stage (Routing, Virtual Channel Allocation, Switch Allocation, Switch Traversal, Link Traversal) consumes exactly one clock cycle.
- **Interaction:** PEs inject packets based on statistical arrival distributions (e.g., Poisson process) at specific clock edges, while routers arbitrate conflicts using round-robin scheduling.

E. Model of Computation

The primary Model of Computation (MoC) for this project is **Discrete-Event Simulation (DES)** combined with a **Synchronous Reactive** model.

- **Discrete-Event:** Packet generation, fault injection, and end-to-end latency calculations are triggered as discrete events in the simulation timeline.
- **Synchronous Reactive:** The internal mechanics of the NoC (routers, arbiters, links) react synchronously to a global clock. At each tick, the state of the buffers and links is updated.

This MoC perfectly aligns with the Noxim C++ architecture (SystemC based), which is explicitly designed to simulate synchronous digital hardware through discrete events.

F. Assumptions and Design Limitations

Assumptions:

- Individual chiplets are assumed to employ locally deadlock-free routing algorithms (e.g., XY routing) prior to integration.
- Faults are assumed to be permanent (e.g., broken microbumps) rather than transient.
- At least one VL per chiplet remains functional.

Limitations:

- The VL selection optimization is performed offline; the boundary routers do not recalculate paths dynamically during runtime but rely on the pre-loaded lookup tables.

G. Resource Model

The simulation restricts the available hardware resources to reflect a realistic 2.5D system constraint:

- **Memory/Buffers:** Each input port of a router is equipped with a limited buffer depth of exactly 4 flits per Virtual Channel.
- **Virtual Channels:** The system is strictly limited to 2 VCs (one for VN.0 and one for VN.1) to minimize area footprint.
- **Bandwidth:** Links are modeled with a standard 32-bit flit width, processing one flit per clock cycle.

These tight buffer and VC constraints make deadlock avoidance highly challenging, thus rigorously validating the effectiveness of the DeFT algorithm.

V. Methodology

This section describes the development and integration of the proposed routing framework. A clear distinction is made between the baseline simulation infrastructure and the custom logic implemented for this project.

Reused Components: The cycle-accurate NoC simulation environment relies on the open-source Noxim framework [2]. Noxim natively provides the hardware modeling of processing elements, routers, crossbars, arbiters, and flit-level credit-based flow control. Standard baseline routing algorithms, such as Dimension-Order Routing (XY), are also reused for intra-chiplet traffic comparison.

Implemented by the Author: The core contributions implemented in C++ specifically for this project include:

- 2.5D topology generation (connecting isolated chiplet meshes to the interposer layer via Vertical Links).
- The DeFT inter-chiplet routing protocol and Virtual Network (VN) state machine.
- The offline optimization script to generate VL Selection Lookup Tables (LUTs).
- A centralized Fault Injection Manager to systematically disable VLs.

A. Implementation Details (Implementation Track)

Control Logic and Routing Tables: The routing logic is split into intra-chiplet and inter-chiplet phases. For intra-chiplet communication, packets strictly follow the deadlock-free XY routing algorithm. For inter-chiplet communication, boundary routers utilize pre-computed Lookup Tables (LUTs). Instead of computing paths dynamically, the offline VL-selection algorithm evaluates all possible fault scenarios and generates optimal mappings. These LUTs are loaded into the boundary routers' memory during initialization. When a packet arrives, the router queries the LUT using the destination chiplet ID and the current fault vector to determine the optimal exit VL.

Fault Injection Control: Fault injection is controlled via a centralized simulation configuration file. Before the first clock cycle, the *Fault Injection Manager* reads the configured fault rate (e.g., 25%) and pseudo-randomly marks specific VL ports as "offline". The boundary routers read this global fault state to activate the corresponding fallback routing table from their LUT.

Communication Mechanisms: Flit transmission relies on Noxim's default wormhole switching and credit-based flow control. However, to support DeFT, the physical links are multiplexed into two distinct Virtual Channels (VCs), representing VN.0 and VN.1.

B. Algorithm Description

The methodology relies on two primary algorithms: VN Assignment (Algorithm 1) to guarantee deadlock-freedom, and VL Selection (Algorithm 2) to maintain fault tolerance and minimize latency.

1) Virtual Network (VN) Assignment: To prevent cyclic dependencies across the "up-down-up" interposer routing, packets are dynamically assigned to one of two Virtual Networks (VN.0 or VN.1) according to three strict rules [1]. Transitions from VN.1 back to VN.0 are strictly forbidden. The assignment logic evaluated at each router is presented in Algorithm 1.

if Current router is Source **then**
    if Source $\in$ {interposer $\cup$ dest. chiplet $\cup$ boundaries} **then**
        Round-robin assignment between VN.0 and VN.1
    else {Destination is on a different chiplet}
        Assign VN.0
    end if
else if Current router $\in$ boundary routers **then**
    if going to the interposer **then**
        Round-robin reassignment between VN.0 and VN.1
    else if coming from the interposer **then**
        Go to (remain in) VN.1
    end if
else
    Stay in the previously assigned VN
end if

Fig. 2. Pseudocode for DeFT Virtual Network (VN) Assignment

2) Fault-Tolerant VL Selection: The VL selection is performed offline to generate the LUTs. It utilizes an exhaustive search optimization to find a selection set (s^*) that minimizes a cost function (C_s). The cost function balances the physical Manhattan distance (D_v) and the traffic load imbalance (L_v) across available VLs.

Require: Set of functional VLs (VL_C), Traffic injection rates
Ensure: Optimal VL selection set s^*
for each s in all possible selection sets S **do**
    for $v \in VL_C$ **do**
        Calculate load cost L_v based on Eq. 2
        Calculate distance cost D_v based on Eq. 4
    end for
    Calculate overall cost $C_s = \sum_{v \in VL_C} (\rho \times D_v) + L_v$
    if $C_s < C_s^*$ **then**
        $C_s^* \leftarrow C_s$
        Update saved selection sets $s^* \leftarrow s$
    end if
end for

Fig. 3. Pseudocode for Offline Congestion-Aware VL Selection

VI. Experimental Setup

The evaluation of the proposed DeFT implementation is conducted through rigorous cycle-accurate simulations. The experiments are designed to benchmark the algorithm's fault tolerance, latency, and throughput under varying network conditions.

A. Platform and Tools

The primary evaluation platform is the **Noxim** Network-on-Chip simulator [2], which provides cycle-accurate modeling of routers, links, and virtual channels. Noxim is extended in C++ to support 2.5D active interposer topologies and custom Virtual Network (VN) assignment logic. For real-world application traffic generation, the **GEM5** full-system simulator is utilized to extract memory traces from a simulated 64-core x86 architecture. All simulations and offline VL-selection optimizations are executed on a Linux-based environment (e.g., Ubuntu 22.04).

B. Datasets and Workloads

The system is evaluated using both synthetic traffic patterns and real-world workloads to ensure comprehensive testing:

- **Synthetic Traffic:**
  - *Uniform:* Packets are uniformly distributed across all destination nodes.
  - *Localized:* 40% of the generated packets are destined for nodes within the same source chiplet, mimicking high spatial locality.
  - *Hotspot:* 3 specific nodes are designated as hotspots, each receiving an additional 10% directed traffic rate.
- **Real-Application Traffic:** Memory traces are extracted from the **PARSEC benchmark suite** [10], specifically utilizing workloads such as *blackscholes*, *fluidanimate*, and *swaptions* to model realistic multi-core communication behavior.
- **Fault Injection Scenarios:** Vertical Link (VL) faults are statically injected at the start of each simulation run. The fault rate ranges from 3.125% (1 faulty VL) up to 25% (8 faulty VLs) to observe the degradation and recovery capabilities of the routing logic.

C. Evaluation Metrics and Baselines

To isolate the improvements provided by the DeFT logic, the performance is compared against standard baseline configurations.

Baselines for Comparison:

- **XY Routing (Fault-Free):** Standard dimension-order routing in a 0% fault scenario to establish the absolute maximum performance upper bound.
- **XY Routing (Fault-Injected):** Standard routing applied to the faulty topology to demonstrate baseline failure rates and deadlocks.
- **Literature Baselines:** Conceptual comparisons will be drawn against the Modular Turn Restriction (MTR) [3] and Remote Control (RC) [9] algorithms based on their theoretical fault-tolerance limits.

Evaluation Metrics: The following metrics are collected from the Noxim simulation logs at the end of each execution cycle:

1) **Reachability (%):** The ratio of successfully delivered packets to the total number of injected packets. This is the primary metric for evaluating fault tolerance; a resilient system should maintain 100% reachability as long as the chiplets are not physically isolated.
2) **Average Latency (cycles):** The average time taken for a packet to traverse from its source Processing Element (PE) to its destination PE, including buffering delays and router pipeline stages.
3) **Network Throughput (flits/cycle/IP):** The maximum rate at which the network can deliver data per active IP node before reaching saturation.

VII. Project Timeline and Milestones

To ensure the systematic execution of this implementation within the semester boundaries, the project workflow is divided into five primary phases. This timeline directly addresses the milestones suggested in the preliminary proposal feedback, transitioning from the theoretical foundation to empirical evaluation.

TABLE III
PROJECT TIMELINE AND MILESTONES FOR THE SPRING SEMESTER

Phase	Tasks & Milestones	Target
Phase 1	**Literature & Architecture:** Define 2.5D topology, system models, and review NoC routing algorithms.	Completed
Phase 2	**Simulator Extension:** Extend Noxim to support 2.5D networks and implement VL fault injection mechanisms.	Weeks 8-9
Phase 3	**Algorithm Implementation:** Code DeFT routing logic, VN assignment, and generate offline selection LUTs.	Weeks 10-11
Phase 4	**Baseline & Evaluation:** Implement baseline XY routing. Run simulations under various traffic loads and fault rates.	Weeks 12-13
Phase 5	**Analysis & Final Report:** Evaluate reachability, latency, and throughput metrics. Finalize the project report.	Week 14

VIII. System-Level Considerations

As a simulation-based study focusing on 2.5D network routing, several system-level design implications must be explicitly addressed. While the primary focus of this implementation is reliable packet delivery, the broader impacts on system performance and physical hardware constraints are critical. For this extended proposal, the following considerations highlight both the directly implemented features and the theoretical limitations that will be thoroughly analyzed in the final report.

Implemented and Evaluated Features:

- **Reliability, Fault Tolerance Limits, and Deadlock Guarantees:** System reliability is the core implemented feature. Deadlock avoidance is strictly guaranteed through the implemented 3-rule Virtual Network (VN) assignment logic. The theoretical fault tolerance limit is bounded only by complete physical disconnection; the simulation will evaluate the system up to a severe 25% Vertical Link (VL) failure rate to demonstrate 100% reachability.
- **Latency and Throughput under Fault Conditions:** Real-time constraints in NoCs are heavily dependent on latency bounds. The implementation actively measures average network latency and throughput degradation. The congestion-aware VL selection logic aims to satisfy timing constraints by proactively routing traffic away from heavily loaded or faulty links.

Discussed Limitations and Future Work:

- **Energy Efficiency and Resource Overhead:** Implementing DeFT requires maintaining two Virtual Channels (VCs) across the network and storing offline Lookup Tables (LUTs) in the boundary routers. While not physically synthesized in silicon for this project, the theoretical power and area overhead of these additional resources will be discussed as a limitation (conceptually $< 2\%$ according to baseline literature [1]).
- **Scalability with Increasing Chiplet Count:** As the number of integrated chiplets increases in future architectures (e.g., scaling from 4 to 16 chiplets), the computational complexity of the offline VL selection optimization and the memory footprint required for the LUTs will grow. While the current simulation model is bounded to 4 chiplets, scalability implications will be critically analyzed.
- **Security Considerations:** Data security is outside the scope of the current routing implementation. However, in heterogeneous 2.5D systems, inter-chiplet communication via an active interposer is vulnerable to side-channel attacks and hardware Trojans. Securing the DeFT routing protocol against denial-of-service or packet interception is designated strictly as future work.

Acknowledgments

I would like to express my sincere gratitude to the course instructor, Assist. Prof. Dr. Selma Dilek, for her valuable feedback and for encouraging the alignment of this project with my Master's thesis. Her guidance significantly helped refine the technical scope and methodology of this proposal. I also extend my deepest thanks to my thesis advisor, Prof. Dr. Süleyman Tosun, Head of the Computer Engineering Department. His continuous support and supervision during

my research on 2.5D chiplet networks provided the foundation for this work, which serves as a baseline for my future studies on reinforcement learning-based routing.

References

[1] E. Taheri, S. Pasricha, and M. Nikdast, "Deft: A deadlock-free and fault-tolerant routing algorithm for 2.5d chiplet networks," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, 2022, pp. 1047–1052.

[2] V. Catania *et al.*, "Cycle-accurate network on chip simulation with noxim," *ACM Transactions on Modeling and Computer Simulation (TOMACS)*, 2016.

[3] J. Yin *et al.*, "Modular routing design for chiplet-based systems," in *Proceedings of the 45th Annual International Symposium on Computer Architecture (ISCA)*, 2018.

[4] X. Wang *et al.*, "Bup: A deadlock resolution framework for 2.5d chiplet networks by update packet bypassing," in *2023 IEEE 31st Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM)*, 2023.

[5] E. Taheri, M. Nikdast, and S. Pasricha, "Red: A reliable and deadlock-free routing for 2.5-d chiplet-based interposer networks," in *Proceedings of the 41st IEEE International Conference on Computer Design (ICCD)*, 2023.

[6] Y. Li *et al.*, "Fault-tolerant circular routing algorithm for 3d-noc," *The Journal of Supercomputing*, 2020.

[7] R. Salamat *et al.*, "A runtime fault-tolerant routing scheme for partially connected 3d networks-on-chip," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2018.

[8] C. Hsu *et al.*, "Configurable fault-tolerant link for inter-die communication in 3d on-chip networks," in *Proceedings of the 2015 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2015.

[9] P. Majumder *et al.*, "Remote control: A simple deadlock avoidance scheme for modular systems-on-chip," *IEEE Transactions on Computers (TC)*, 2020.

[10] C. Bienia and K. Li, "Benchmarking modern multiprocessors," in *Princeton University technical report*, 2011.